BIG DATA TECHNOLOGIES

INFRASTRUCTURE HEALTH ANALYSIS FOR DIGITAL HOSPITALS USING THE HADOOP ECOSYSTEM

MENTOR: Dr N. Shirisha, Assoc.Prof.

BATCH NO: 19

TEAM MEMBERS:

2320030082 B. Manish

2320030142 BSR Chaitanya Kousik

2320030229 Surya Teja K

2300090287 Aditya

TABLE OF CONTENTS

1. ABSTRACT
2. INTRODUCTION
3. EXISTING SYSTEM
4. PROPOSED SYSTEM
5. OBJECTIVES
6. LITERATURE SURVEY
7. SOFTWARE REQUIREMENTS
8. HARDWARE REQUIREMENTS
9. FUNCTIONAL REQUIREMENTS
10. NON – FUNCTIONAL REQUIREMENTS
11. SYSTEM ARCHITECTURE
12. UML DIAGRAM
13. ACTIVITY DIAGRAM
14. MODULE DESIGN

REVIEW – 2 COMMENTS ADDRESSED

•Clarification: MySQL is not our Big Data storage; it simulates the hospital's existing legacy system.
•The Pipeline Goal: The project demonstrates extracting data *from* limited legacy systems (MySQL) *into* our actual Big Data environment (Hadoop).
•Why Hadoop over MongoDB: MongoDB is an operational NoSQL database (good for live apps). We need a Data Warehouse (Hive on HDFS), which is specifically optimized for massive, historical, batch-analytical queries.

ABSTRACT

Modern hospitals generate millions of machine-generated logs every hour from Wi-Fi routers, servers, and digital medical devices such as MRI and X-ray systems. Due to their volume and velocity, these logs cannot be efficiently analyzed using traditional database systems. This project implements a Hadoop-based Big Data framework to store and analyze these digital footprints. Apache HDFS is used for scalable and fault-tolerant storage, while Apache Hive enables large-scale analytical querying. The system identifies security threats such as brute-force attacks and detects abnormal machine error patterns, helping improve both network security and operational reliability in hospital environments.

INTRODUCTION

The Problem:
Hospitals are high-value targets for cyberattacks such as ransomware. The massive volume of system-generated logs makes manual monitoring and analysis impractical.

The Data:
The project focuses on analyzing system logs, which are automatically generated text files produced by servers, network devices, and medical equipment.

The Solution:
A distributed log analysis framework using Hadoop. Instead of processing logs on a single machine, the system uses a cluster-based approach to detect anomalous patterns across large datasets.

Objective Overview:
To build a scalable pipeline that ingests, cleans, stores, and queries system logs to identify security threats and operational issues.

EXISTING SYSTEM

Current Method:

- Hospitals use traditional **RDBMS** (MySQL/Oracle) for log storage.
- Security relies on **Firewalls** and real-time **IDS**.
- Data retention is short (24–48 hours) due to storage limits.

Major Drawbacks:

- **Storage Limit:** Traditional databases crash or slow down under terabytes of log data.
- **Security Gap:** Firewalls miss "**Low and Slow**" attacks spread over weeks.
- **No Correlation:** Cannot link logs from Wi-Fi, Servers, and MRI machines into one timeline.
- **Reactive Maintenance:** Equipment is fixed only **after failure** because error patterns aren't analysed.

PROPOSED SYSTEM

- Our goal is to build a centralized **Data Warehouse** using Hadoop to store and analyze all hospital machine logs.
- Main purpose is to move from reactive security to **Proactive Forensic Analysis**.
- We are using **Apache Sqoop** to automate the ingestion of data from diverse sources like VPNs, Routers, and MRI machines.
- We deploy the data into **HDFS** to provide a scalable, fault-tolerant storage environment for terabytes of data.
- **Apache Hive** allows us to query 30 days of history at once to detect "low and slow" hacking patterns.
- This solution helps the hospital identify **Predictive Maintenance** needs by analyzing error trends in medical equipment.
- Utilizing this architecture will allow the hospital to maintain a **tamper-proof audit trail** for legal and regulatory compliance

OBJECTIVES

- To store and process large hospital log datasets reliably using HDFS.
- To automate data cleaning and preprocessing using MapReduce.
- To identify security threats and abnormal access patterns via Hive.
- To establish a reliable, end-to-end data pipeline using Sqoop.
- To improve operational reliability by analysing medical equipment error logs.

SOFTWARE REQUIREMENTS

Below are the software requirements for proposed system

- **Virtualization Tool:** Oracle VM VirtualBox.
- **Hadoop Environment:** Cloudera QuickStart VM (CDH - Cloudera Distribution including Hadoop).
- **Guest Operating System:** CentOS (pre-configured inside the Cloudera VM).
- **Core Framework:** Apache Hadoop (HDFS and MapReduce) for storage and processing.
- **Data Ingestion:** Apache Sqoop for importing raw datasets.
- **Data Warehouse:** Apache Hive for writing analytical queries.
- **Relational Database:** MySQL (built into Cloudera) to simulate the hospital's local database.

HARDWARE REQUIREMENTS

The hardware requirements for the proposed system include the following:

- **Processor:** Intel Core i5 / AMD Ryzen 5 (or higher).
- **RAM (Memory):** Minimum 8 GB RAM.
- **Storage Space:** Minimum 50 GB of free disk space.
- **System Architecture:** 64-bit Host Operating System (Windows / Mac / Linux).
- **Network:** Standard Ethernet or Wi-Fi adapter for internal VM networking.

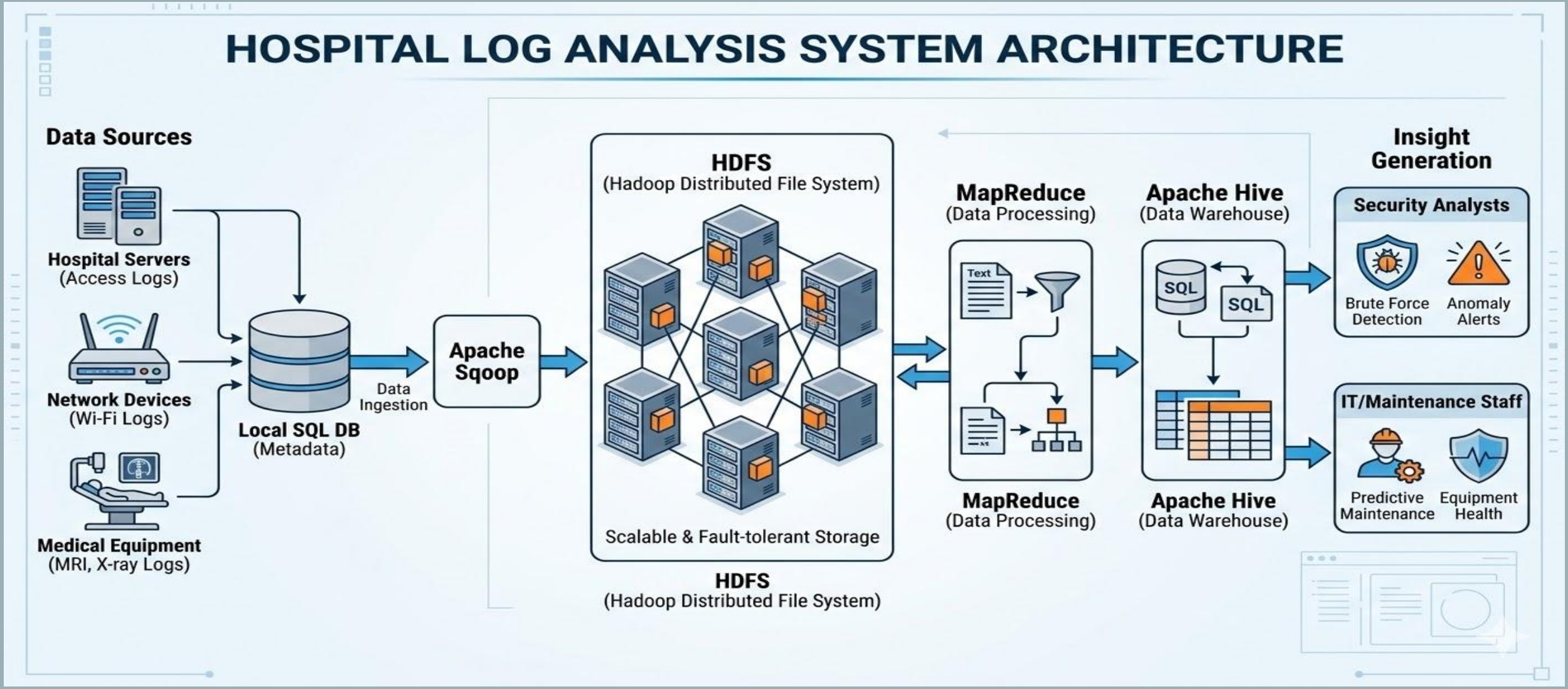
LITERATURE SURVEY

S.No	Research Paper (Title & Link)	Author(s)	Drawbacks / Gaps	Future Directions
1	**Big Data for Secure Healthcare System: A Conceptual Design** https://link.springer.com/article/10.1007/s40747-017-0040-1	Research Team (Secure Healthcare Study)	Conceptual model only; no Hadoop ecosystem implementation.	Implement full Hadoop pipeline (Sqoop → HDFS → MapReduce → Hive).
2	**Log Analysis with Hadoop MapReduce** https://dergipark.org.tr/en/pub/ject/article/980060	Gligor Risteski et al.	Uses traditional MapReduce; high latency for alert generation.	Replace batch processing with Hive/Spark SQL for faster querying.
3	**Enhanced Secured MapReduce Layer for Big Data Privacy and Security** https://doi.org/10.1186/s40537-019-0193-4	P. Jain, M. Gyanchandani, N. Khare	Focused on framework-level security; lacks practical log-analysis case study.	Integrate security layer with healthcare log analytics pipeline.
4	**Digital Forensics Readiness in Big Data Networks: A Framework for Linux–Hadoop Environments** https://doi.org/10.3390/asi7050090	Cephas Mpungu et al.	Emphasizes forensic response after attack, not proactive threat detection.	Develop proactive anomaly detection queries using Hive.
5	**A Novel Hadoop Security Model for Addressing Malicious Collusive Workers** https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8285196/	(PMC Research Group Authors)	Focused only on cluster-node security, not log-based intrusion detection.	Combine node-level protection with log anomaly detection.

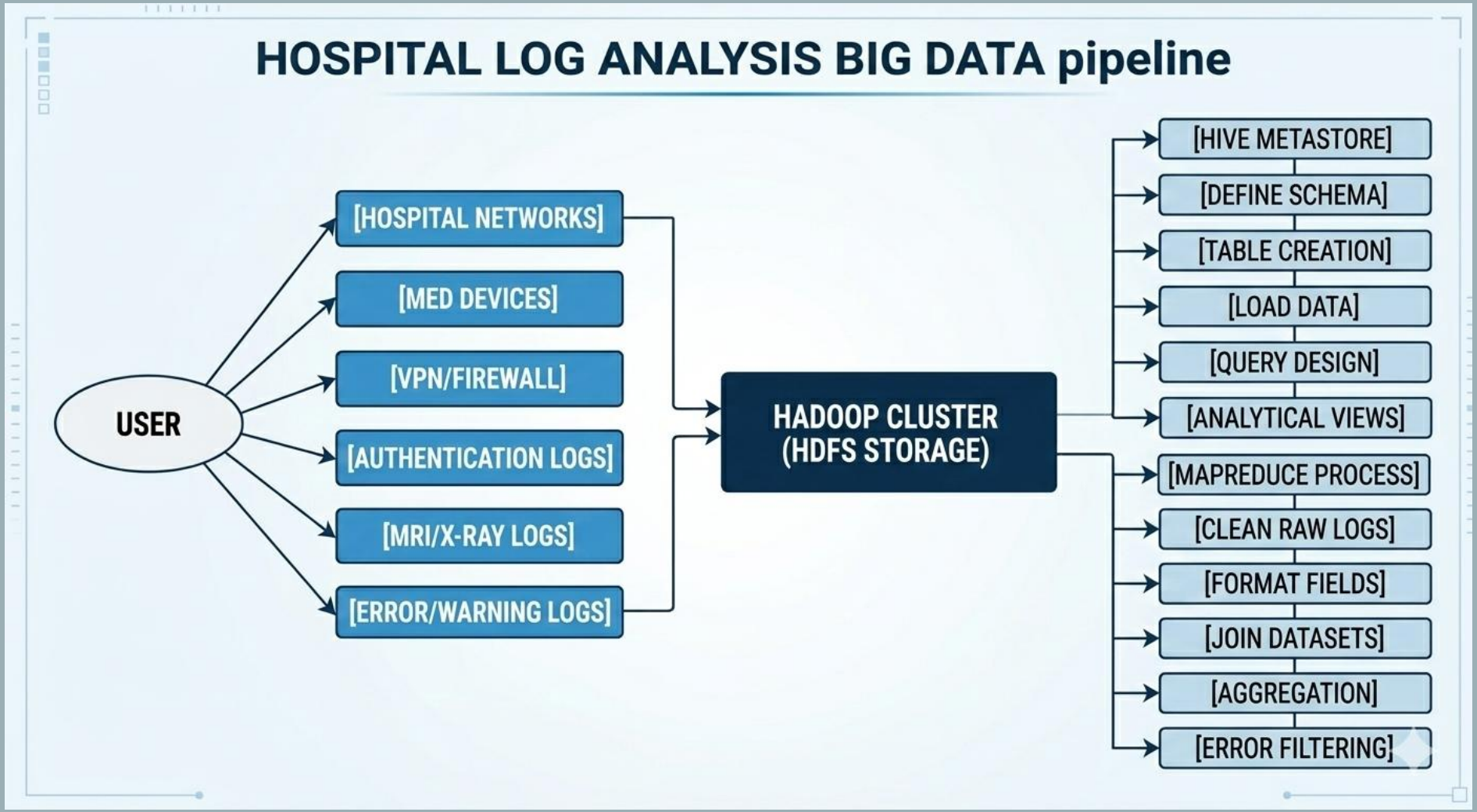
LITERATURE SURVEY

6	**Evading Cyber-Attacks on Hadoop Ecosystem: ML-Based Security-Centric Approach** https://doi.org/10.3390/info15090558	Neeraj A. Sharma et al.	Heavy ML dependency; requires high computational resources.	Implement lightweight rule-based Hive queries for smaller hospitals.
7	**Aadhaar Data Analysis Comparison in MapReduce, Hive and Spark** https://doi.org/10.2991/ahis.k.210913.036	R. Roopa et al.	Not security-focused; dataset unrelated to cybersecurity logs.	Apply comparative analysis specifically to hospital system logs.
8	**Big Data Analytics in Healthcare: Data Sources, Tools, Challenges and Opportunities** https://doi.org/10.1186/s43067-024-00190-w	Multiple Authors (Springer Open Survey)	Broad survey; lacks technical implementation details.	Develop practical Hadoop-based implementation for hospital infrastructure.
9	**Analyzing Web Access Logs using Spark with Hadoop** https://ijcaonline.org/archives/volume180/number1/28768-2017915904/	Vandita Jain et al.	Focused on web logs; not healthcare or ransomware detection.	Extend approach to medical device and network router logs.
`	**Big Data for Secure Healthcare System: A Conceptual Design** https://link.springer.com/article/10.1007/s40747-017-0040-1	Research Team (Secure Healthcare Study)	Conceptual model only; no Hadoop ecosystem implementation.	Implement full Hadoop pipeline (Sqoop → HDFS → MapReduce → Hive).

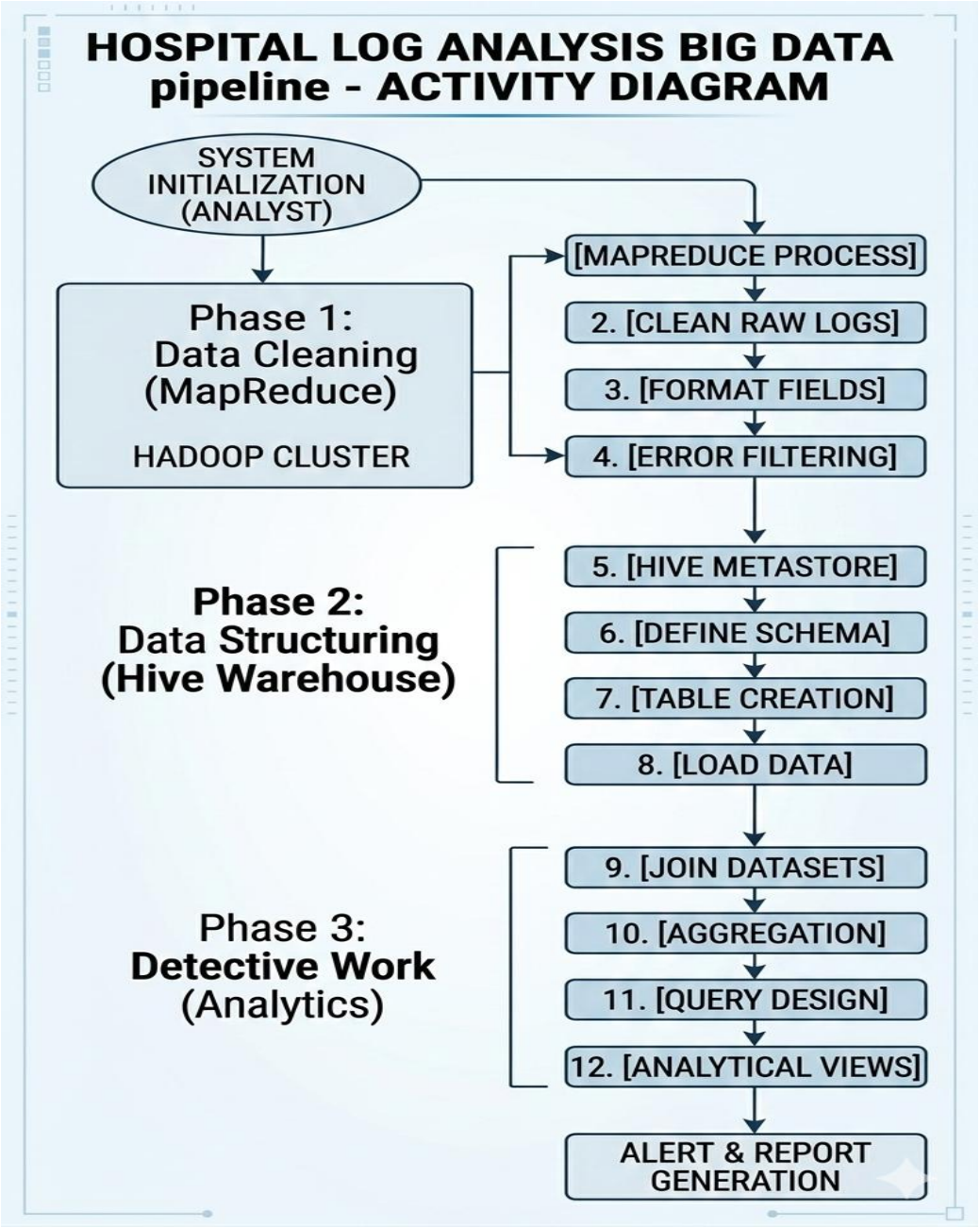
SYSTEM ARCHITECTURE



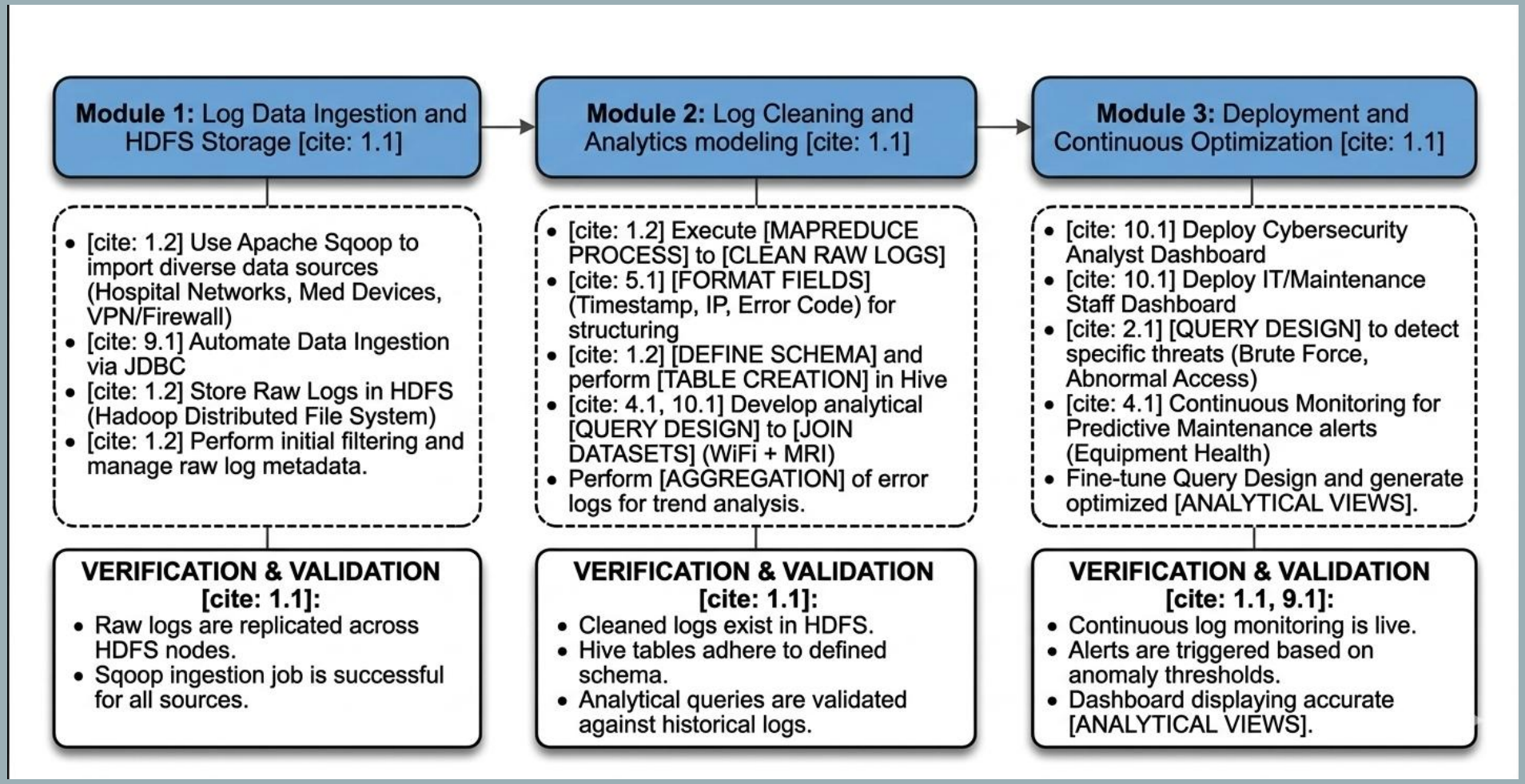
USE CASE DIAGRAM



ACTIVITY DIAGRAM



MODULE DESIGN



THANK YOU